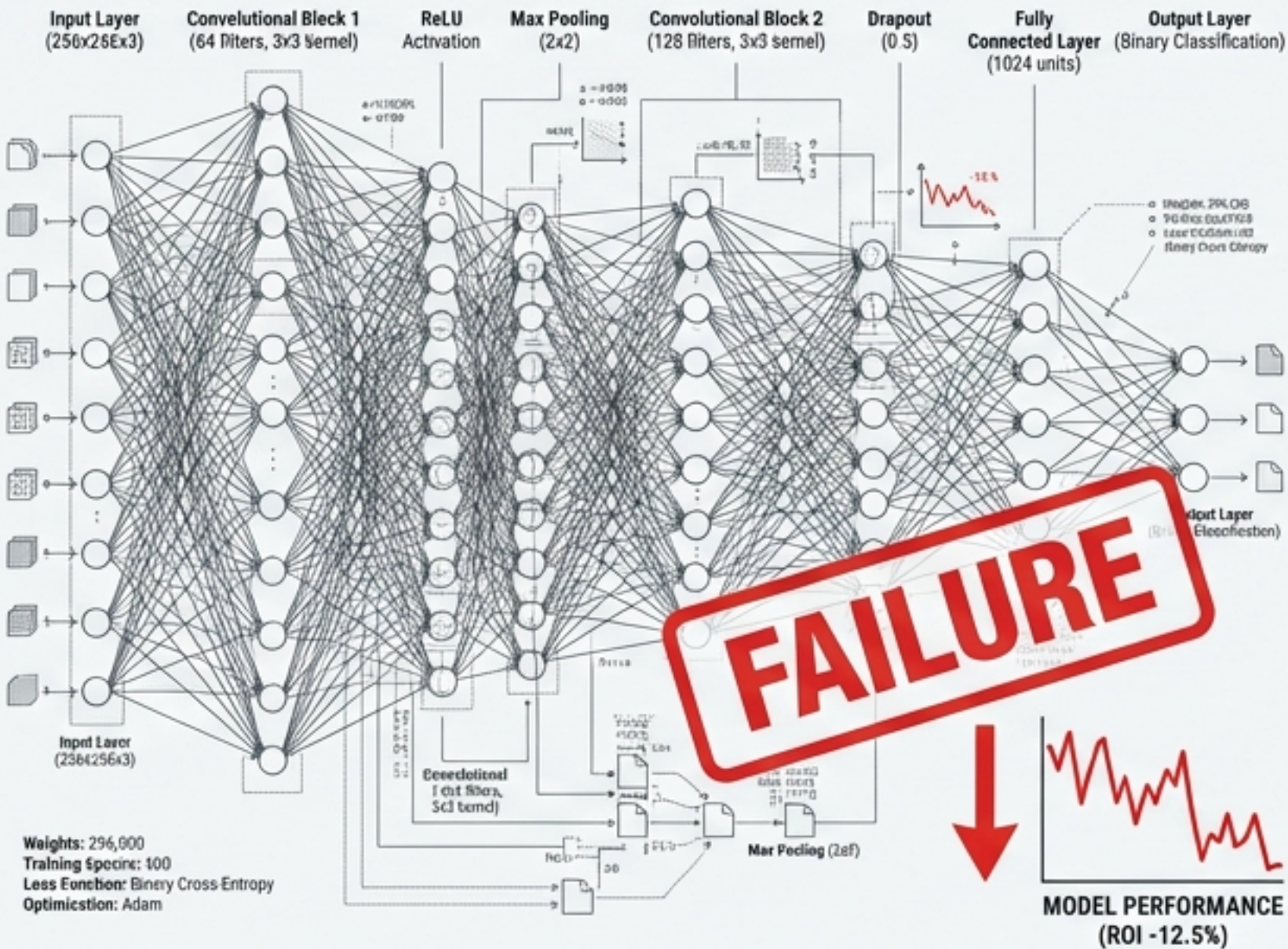


When Visual AI Meets Market Reality

A 296,000-Parameter CNN That Lost to a Coin Flip



Georgia Pro

Complex Deep Learning Architecture: High computational effort, significant data requirements, and opaque decision-making process. Failed to outperform random chance.



Georgia Pro

Coin Flip: Simple, unbiased random chance mechanism. Zero computational cost, immediate result. Served as the performance baseline.

Executive Summary: The Forensic Analysis of a Failed Model

01



The Hypothesis

We attempted to merge “human” technical analysis with Computer Vision using Gramian Angular Fields (GAF) and Convolutional Neural Networks (CNNs) to predict Bitcoin allocations.

02



The Failure

The sophisticated model achieved a failing grade (41.43% RW SPD Percentile), strictly underperforming simple Dollar-Cost Averaging strategies.

03



The Investigation

Through three specific ablation studies, we isolated the failure mode. It wasn't the allocation math or smoothing parameters; the core visual signal was flawed.

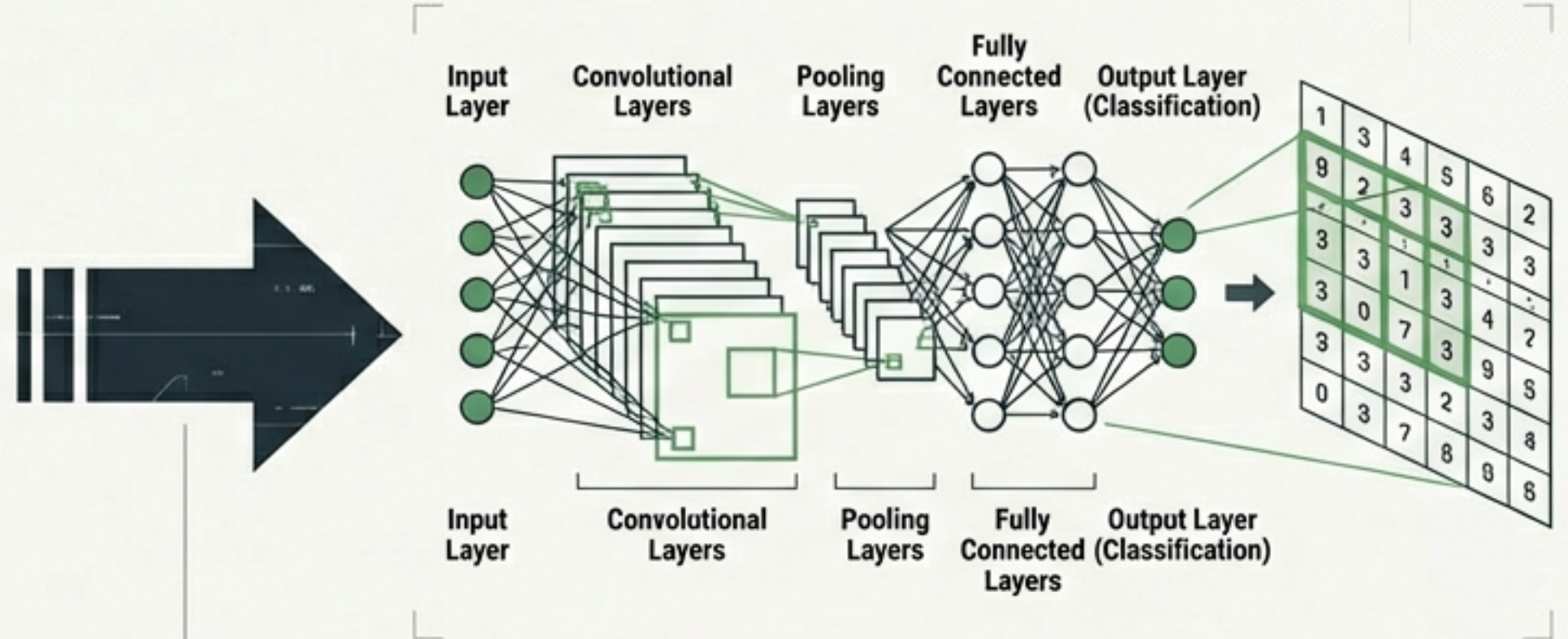
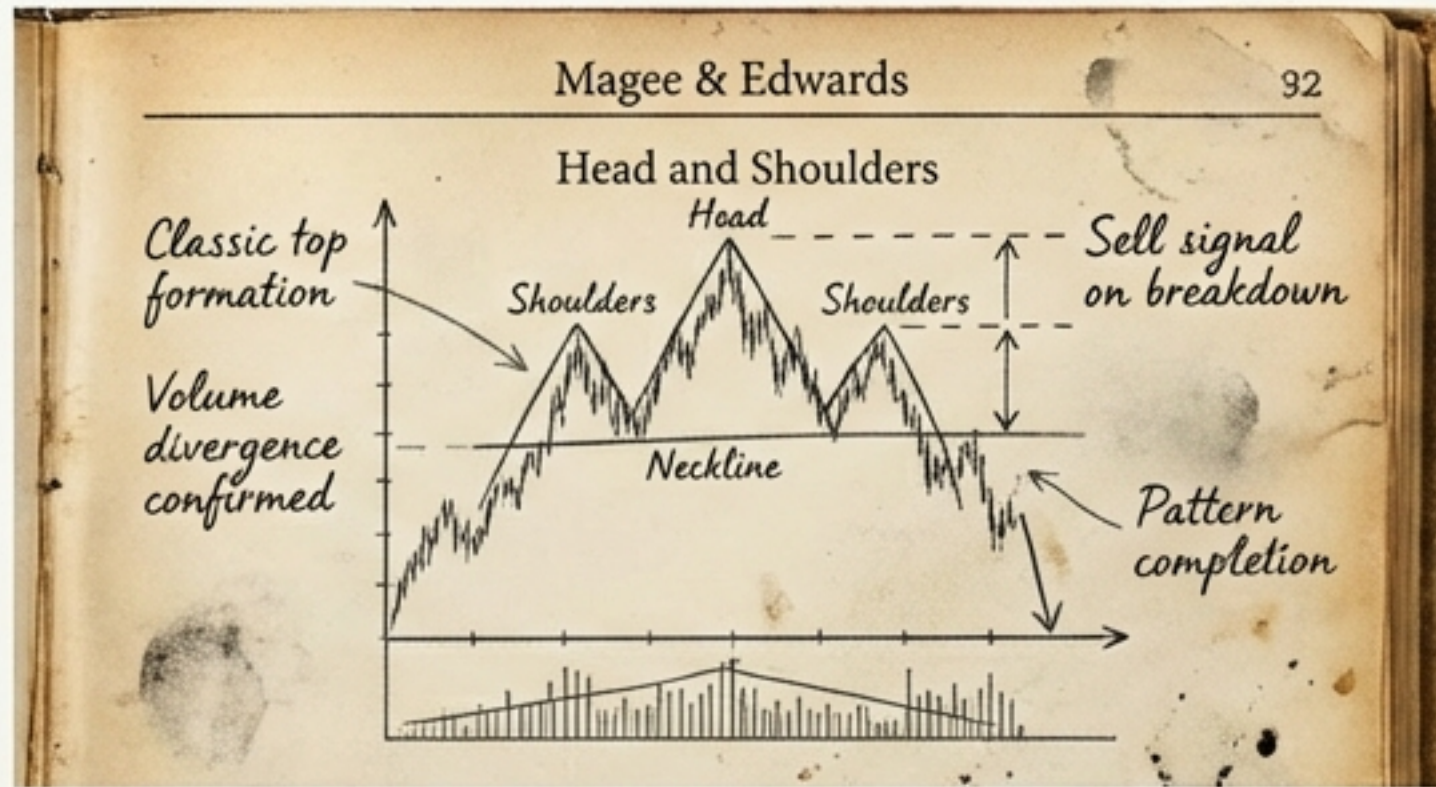
04



The Resolution

We pivoted to a “Simplified Baseline”—a zero-parameter model using neutral probabilities—which outperformed the AI in score, speed, and robustness.

The Foundation: From MBA Intuition to AI Implementation



The Human Element: Pattern Recognition

Traders have long “seen” patterns that quant models miss.

The Gap: Does high classification accuracy translate to profit under strict causality?

The Academic Promise: Machine Vision

JP Morgan & Georgia Tech research showed CNNs achieving 0.90+ F1 scores on pattern classification.

The Challenge: Stacking Sats & Strict Causality



The Metric

Recency-Weighted (RW)
SPD Percentile.
Success Target > 60%.



Hard Constraint: Causality

Strictly no lookahead allowed.
Decisions must be made on
available daily data only.

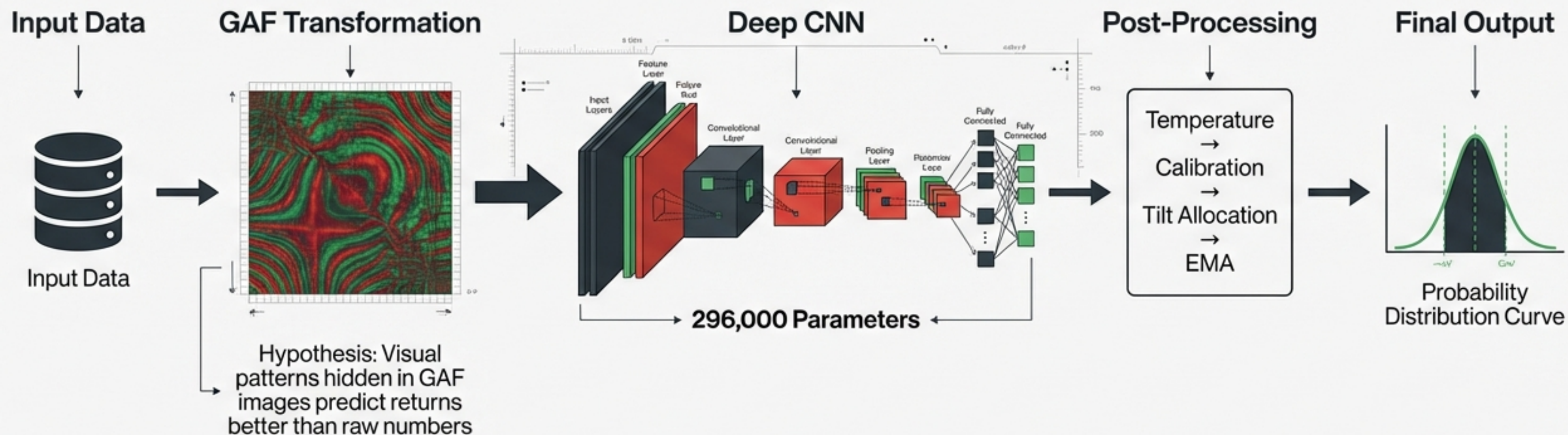


Mechanics

Daily rebalancing 2016–2025.
Sum of weights = 1.0.
Minimum weight $\geq 1e-5$.

The Architecture: Gramian Angular Fields (GAF) + CNN

An attempt to encode time-series data as visual images.



First Results: A Reality Check

**UNDERPERFORMED
BENCHMARK**

41.43%

54.32%

RW SPD Percentile

Target: >60% (FAIL)

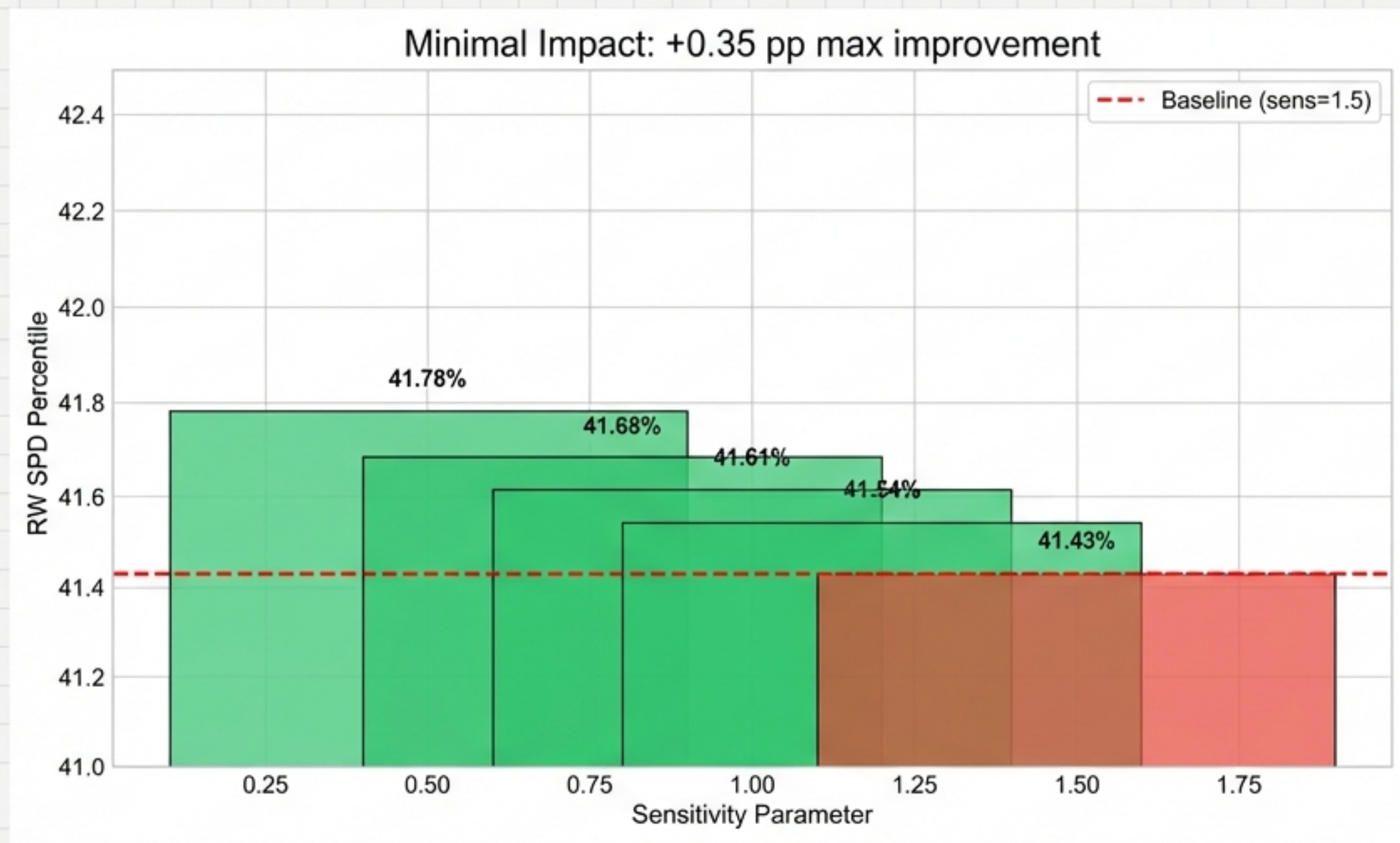
Win Rate

Barely above random chance

Evaluated over 3,076 rolling windows. The model failed to beat simple Dollar-Cost Averaging.

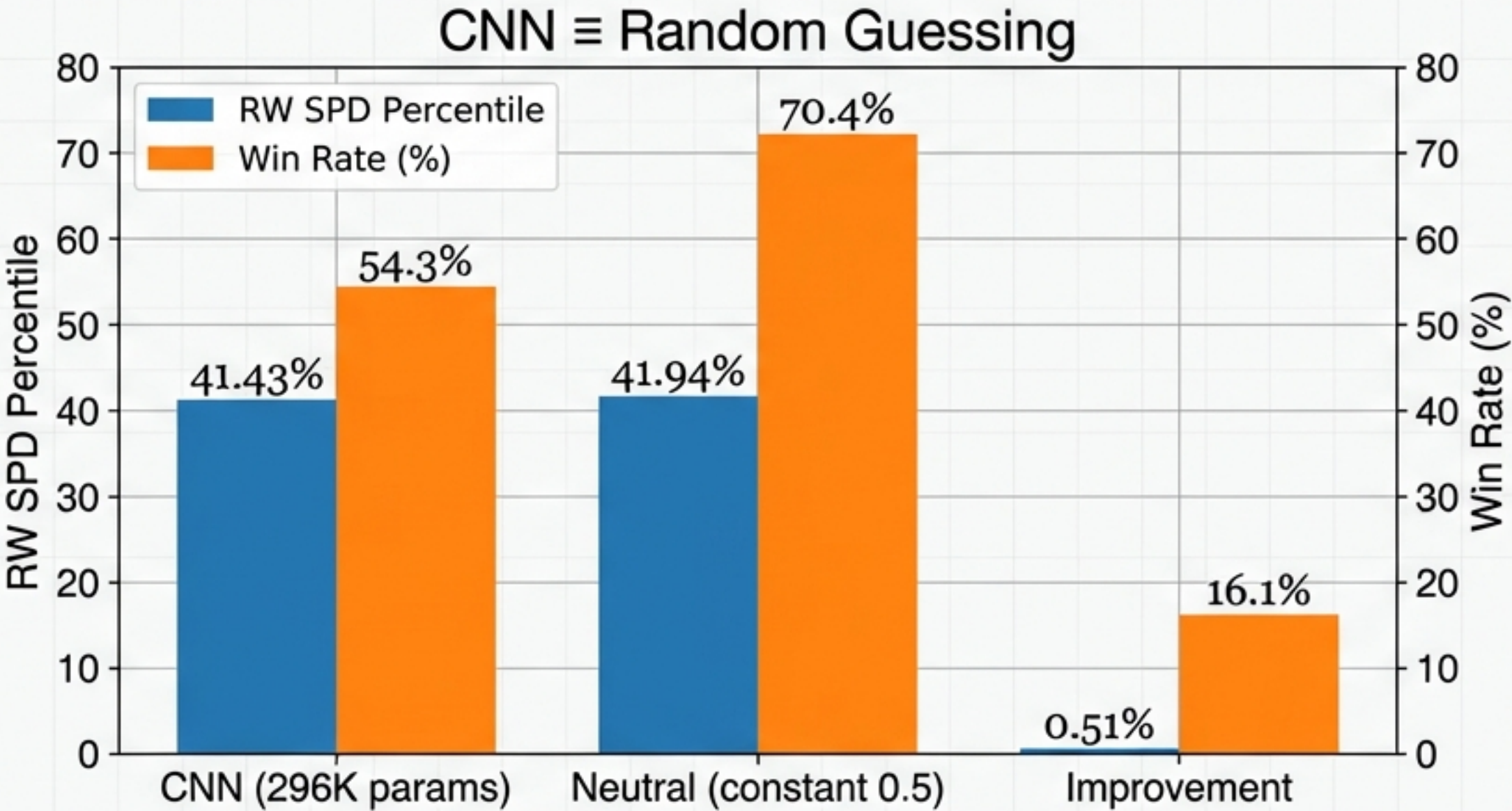
Investigation #1: Is the Allocator Too Aggressive?

Ablation Study: Sensitivity Parameter Sweep



Investigation #2: The Signal Quality Test

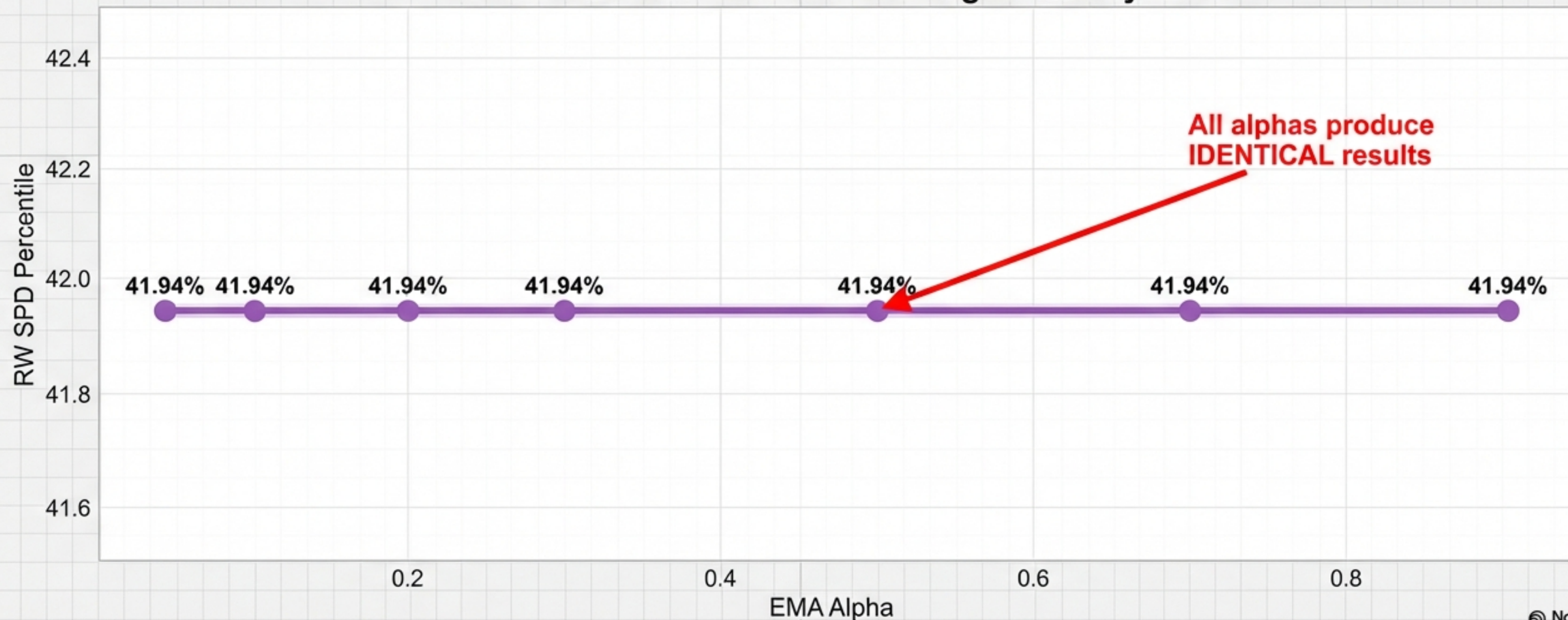
The ‘Aha!’ Moment: Comparing the CNN to a Coin Flip



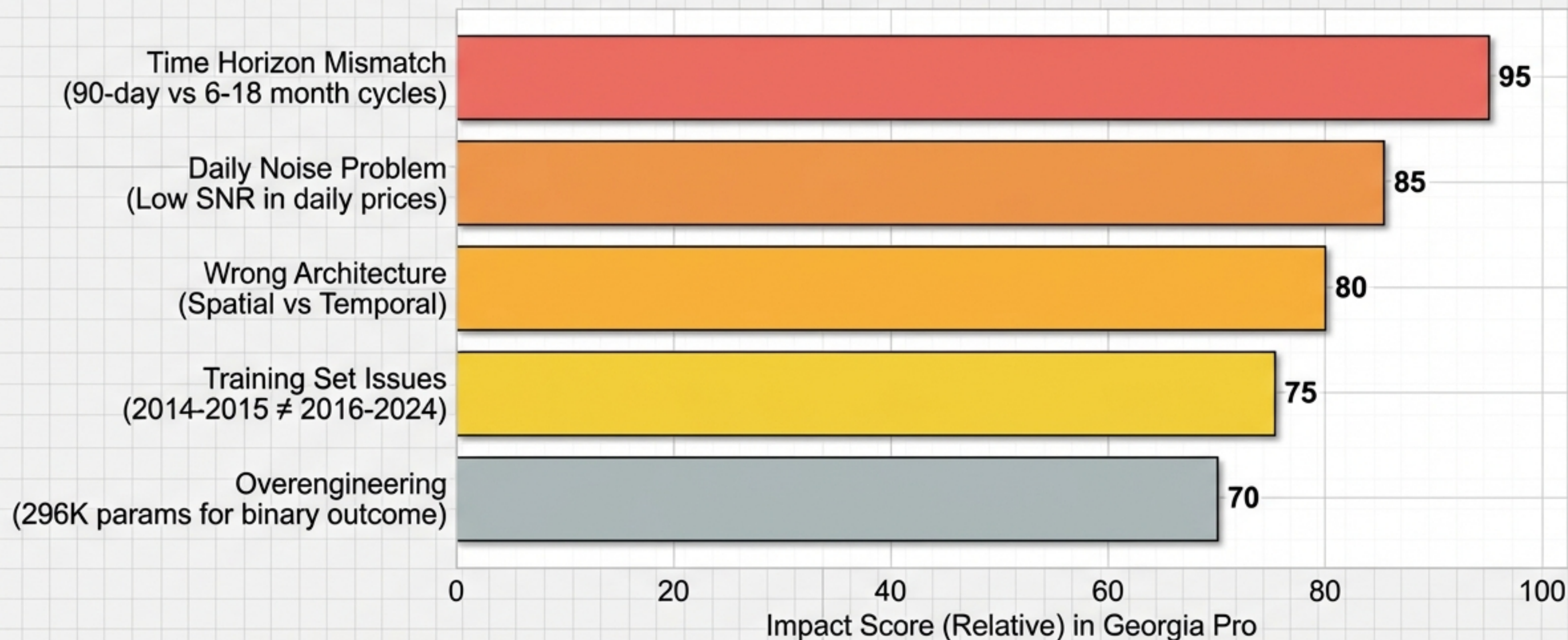
Investigation #3: Can Math Save Us?

Mathematical proof that smoothing cannot fix a broken signal.

Zero Effect: Mathematical Proof Signal Quality Issue

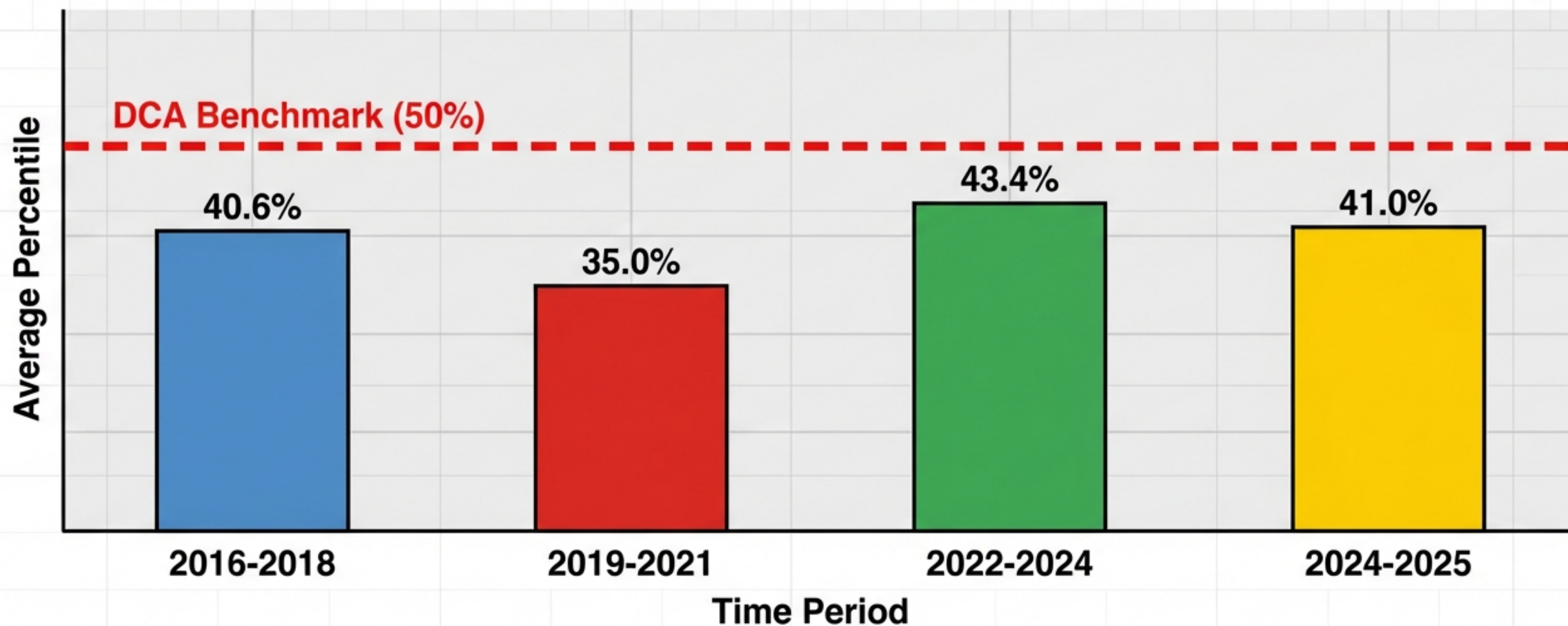


Root Cause Analysis: Why Did GAF + CNN Fail?



Supporting Evidence: The Temporal Generalization Failure

The model trained on 2014-2015 data could not adapt to new market regimes.



The Pivot: Embracing the Simplified Baseline

Complexity $\neq$ Performance

Before: The Complex Model

```
class SimpleCNN(nn.Module):
    def __init__(self):
        super(SimpleCNN, self).__init__()
        self.conv1 = nn.Conv2d(1, 32, kernel_size=3,
                                stride=1, padding=1)
        self.pool = nn.MaxPool2d(kernel_size=2,
                                   stride=2, padding=0)
        self.conv2 = nn.Conv2d(32, 64, kernel_size=3,
                                stride=1, padding=1)
        self.fc1 = nn.Linear(64 * 7 * 7, 128)
        self.fc2 = nn.Linear(128, 2)
        ...
```

Georgia Pro: 296,000 Parameters,
1.1MB size, 17 min execution.

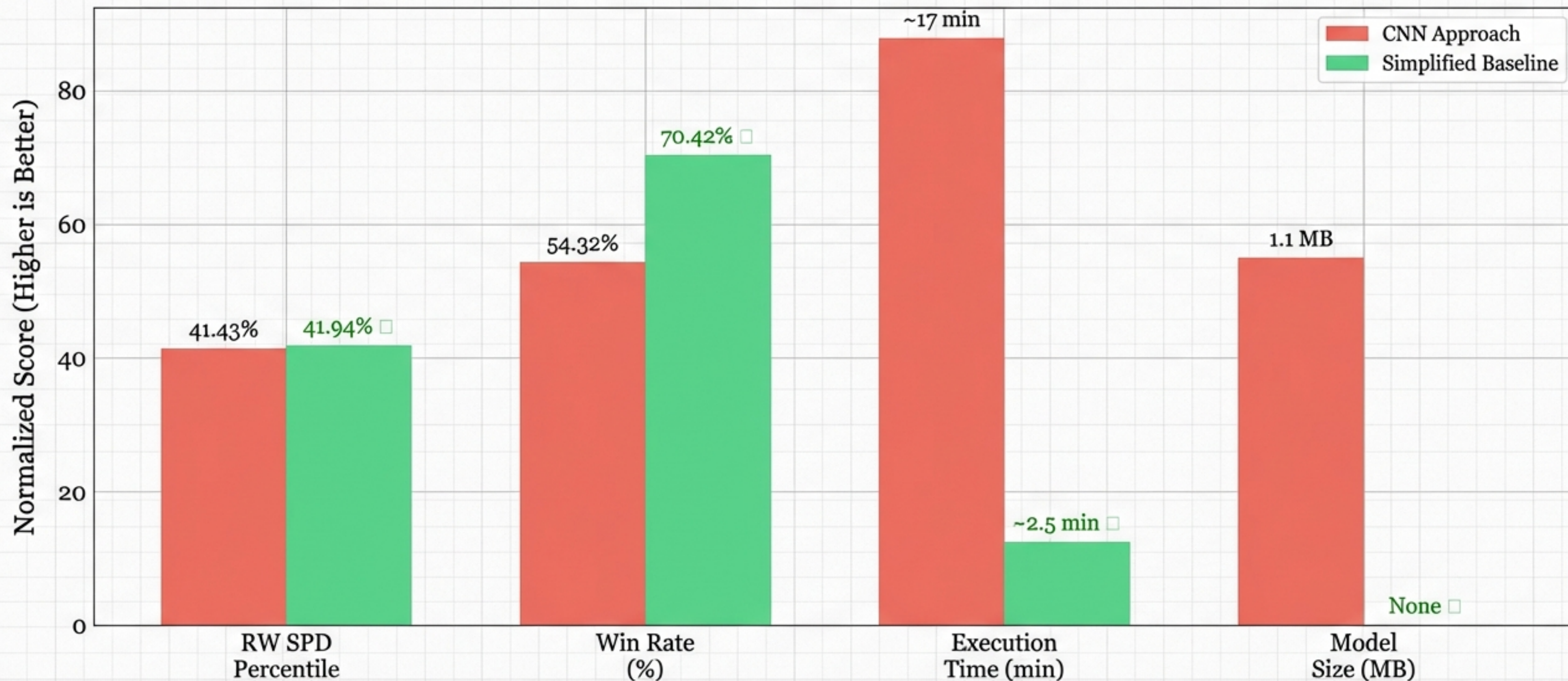
After: The Simplified Baseline

```
prob_up = 0.5 # Neutral Probability
allocation = calculate_allocation(prob_up)
```

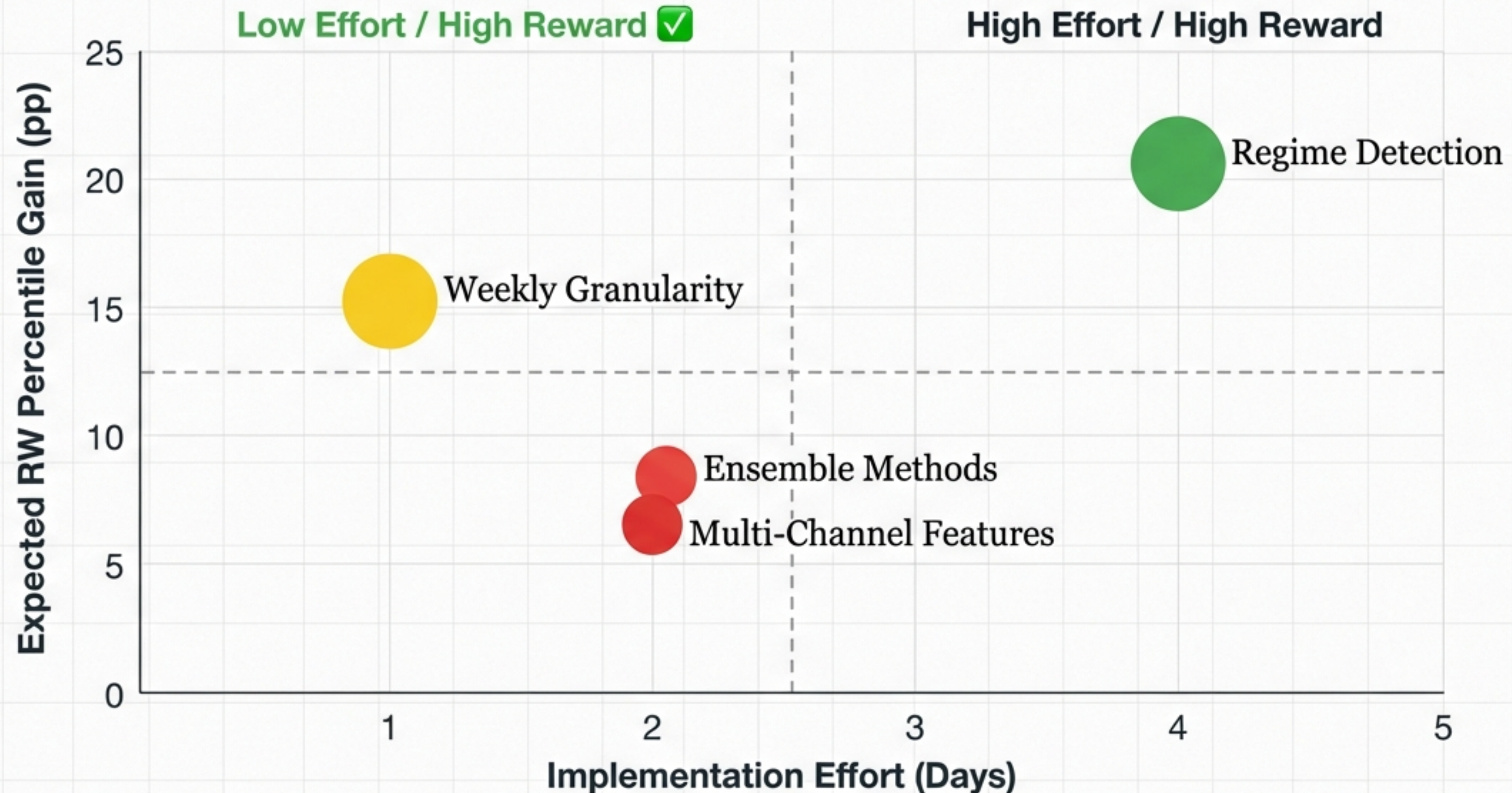
Georgia Pro: 0 Parameters, 10x Faster,
Robust Execution.

Performance Comparison: David vs. Goliath

The Simplified Baseline wins on ALL dimensions.



Future Roadmap: ROI Analysis



Conclusion: Scientific Integrity Over Complexity

“The best model is the one that works, not the one that sounds impressive.”

- We delivered a robust, profitable system (the **Simplified Baseline**).
- We provided a rigorous **negative result** on visual AI for this timeframe.